My Primary School is at the Museum

Research Report

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Table of Contents

1. Research report overview 3

2. Annotated bibliography 4

2.1 Introduction 4

2.2 The policy context 4

2.2.1 Summary 4

2.2.2 Policy context literatures 5

2.3 Cultural capital and socio-economic development 7

2.3.1 Summary 7

2.3.2 Literatures 8

2.4 Sector literatures: museum studies and education studies 9

2.4.1 Summary 9

2.4.2 Sector publications 10

2.4.4 Academic literatures 12

3. Case studies 16

3.1 Summary 16

3.2 Case studies 17

4. Insights and recommendations 31

4.1 Curriculum optimisation 31

4.2 Applied understanding 31

4.3 Other learning 31

4.4 Participant attitudes 32

4.5 Resource management 32

4.6 Recommendations 33

5. List of references 36

1. Research report overview

This report documents the current landscape of museum school partnerships affording school children sustained exposure to museums or other cultural environments and underlines the potential of these so called extended museum school partnerships to effectively deliver curriculum learning whilst also providing additional benefits such as cost efficiencies and non-cognitive learning opportunities.

The report comprises an annotated bibliography providing an overview of academic, "grey" and sector literatures and a case study long list, which collates and describes examples of extended museum school partnerships. Finally, the report compares and contrasts the impacts and outcomes of the extended museum school partnerships included on the case study long list, outlining insights and recommendations for the development of such programmes.

The annotated bibliography sets out the UK policy context for museum school partnerships and explains how extended museum school partnerships have the potential to meet cultural education policy requirements more effectively than basic versions of museum school partnerships. Following this, the report discusses the potential of extended museum school partnerships to address the relationship between low educational attainment and low socio-economic status. The annotated bibliography also presents the way in which extended museum school partnerships have been handled within museum and education studies, noting particularly how extended museum school partnerships can help children to improve their social skills and ability to understand school subjects within real-world contexts.

Finally, the report foregrounds several issues that could influence the advancement of extended museum school partnerships which include: the impacts of extended exposure to museums on children's subject-based learning and other learning; the effective evaluation of learning; collaboration between education communities; and, the role of management strategies such as co-location and co-production as a valuable way of ensuring that children's cultural education is not negatively impacted by reduced financial resources.

2. Annotated bibliography

2.1 Introduction

The first section of this report is an annotated bibliography compiling sources of information relating to the My Primary School is at the Museum project. The sources of information comprise policy, publications from the museum and education sectors, "grey" literature and academic literature. The literatures are organised according to the key areas of thought informing the advancement of extended museum-school partnerships: policy, socio-economic development, and education. Each of these sections opens with a summary before the relevant literatures are listed. It should be noted, however, that though the areas of thought by which this bibliography is organised might be the key drivers of extended museum-school partnerships, the underlying issues informing them are often intertwined.

2.2 The policy context

2.2.1 Summary

Overall, UK Government policy, which requires all school children to receive a minimum level of cultural education, provides a positive context in which to advance extended museum-school partnerships. As successive UK Governments (since at least 1997) have agreed, cultural education is important and museums are well placed to contribute to its delivery. The term cultural education, it should be noted, is broad and relates to many different iterations of culture including literature, dance, music, history, film, theatre and art.

The Henley (2012) report on cultural education in England for the Department for Education (DfE) and the Department of Culture, Media and Sport (DCMS) argues that, whilst cultural education in England is good for some students, it should be improved and increased for all students. In other words, there is a need for students to receive a broad and continuous cultural education, characterised by opportunities to engage in art and cultural practices outside the classroom. These opportunities might include museum or theatre trips, field trips, dance and music lessons or exposure to creative practitioners. In essence, the Henley report argues that good cultural education has the potential to deliver direct educational benefits to children through the acquisition of knowledge, future economic benefits through the development of a well-rounded workforce and social cohesion through the promotion of a shared cultural identity.

To improve and expand the provision of cultural education in the UK, the Henley report recommends increased collaboration between cultural institutions and educational institutions as well as greater and more formalised recognition of cultural activities within the education system. Museums, heritage sites, or other cultural institutions are seen as existing resources, with accompanying expertise, able to deliver, or partially deliver, these recommendations. Indeed, it is possible to see a cascade of activities following recommendations made by the report in the development of regional networks of museum-school partnerships facilitated by publically funded bodies such as the Arts Council England (ACE), the Heritage Lottery Fund (HLF), the British Film Institute and Historic England.

In 2016, the UK Government published the Culture White Paper, which set out their support for equitable public access to cultural experiences. Broadly speaking, this paper agrees with the conclusions and recommendations of the Henley report, adding further impetus to the case for cultural education.

However, as underlined by the Warwick Commission on the Future of Cultural Value, the emphasis placed by the UK Government on the cognitive and social impacts of cultural education on children has perhaps led to the prioritisation of particular types of museum-school partnerships that can be measured according to existing educational evaluative frameworks. Other impacts of museum school partnerships, then, such as non-cognitive learning or creativity, which may not easily correspond to existing evaluative frameworks, are often not adequately represented in the practice of museum-school partnerships (Hall & Thomson, 2007; Neelands, 2015). This risks the development of nominal museum school partnerships that do not deliver enriching cultural experiences. This problem suggests a need for innovative partnerships that focus on the provision and exploration of the expanded impacts of cultural education, rather than on singular outcomes.

In essence, then, there should be no policy barriers to the development of successful museum-school partnerships and, in fact, policy should drive the development of new and existing museum school partnerships. In addition, museum school partnerships based on the principle of extended exposure to culture or on a 'residency' model could not only meet policy requirements but also offer the opportunity to address under-representation of non-cognitive or expanded learning models in museum school partnerships and mitigate the risk of nominal partnerships that fail to offer adequate enrichment.

2.2.2 Policy context literatures

Arts Council England (n.d). *Cultural Education; a guide for Governors*. Retrieved from http://www.artscouncil.org.uk/sites/default/files/download-file/ACE172%20Cultural%20Education%20for%20Governors%20-%20WEB.pdf

- A practical guide to inform school governors of the requirements of schools to provide cultural education.

*Arts Council England (n.d.) *Museums and Schools Programme*. Retrieved from http://www.artscouncil.org.uk/programme/museums-and-schools-programme

- The webpages documenting ACE's museums and schools programme, established in response to the Government recommendation for increased and improved cultural education.

Belfiore, E., & Bennett, O. (2008). *The social impact of the arts: an intellectual history*. Basingstoke: Palgrave Macmillan.

- This book provides an analysis of the power of the arts to afford individual and societal change.

Children, Schools and Families Committee. (2010). *Transforming Education Outside the Classroom*. Retrieved from https://publications.parliament.uk/pa/cm200910/cmselect/cmchilsch/418/418.pdf

- A parliamentary committee report recommending that state-funded schools should provide out-of-classroom learning experiences for all students. The report recommends that adequate funding should be provided and that associated health and safety protocols should be streamlined to assist in this process.

Crossick, G., & Kaszynska, P. (2016). *Understanding the Value of Arts and Culture*. Retrieved from http://www.ahrc.ac.uk/documents/publications/cultural-value-project-final-report:

- This report by the Arts and Humanities Research Council funded project looks at how we value arts and culture in society and on an individual basis.

*Department for Culture, Media and Sport. (2016). *Culture White Paper*. Retrieved from https://www.gov.uk/government/publications/culture-white-paper.

- A report presenting the Government's approach to supporting UK culture, which is based on the provision of equitable access to cultural experiences.

*Department for Education and the Department for Culture, Media and Sport. (2012). *The government response to Darren Henley's review of cultural education*. Retrieved from https://www.gov.uk/government/uploads/system/uploads/attachment_data/file/180845/DFE-00020-2012.pdf

- In response to the Henley review of cultural education in the UK, this report presents the actions taken (or to be taken) by the UK Government in order to meet the recommendations of the report. Importantly, the report sets out the mechanisms and networks through which cultural education is expected to be delivered.

*Henley, D. (2012). *Cultural Education in England*. Retrieved from https://www.gov.uk/government/uploads/system/uploads/attachment_ ata/file/260726/Cultural_Education_report.pdf

- This report, commissioned by the UK Government, presents the importance of good cultural education, looks at the current provision of cultural education in England and sets out measures for its improvement.

Historic England (2017). *Heritage Counts*. Retrieved from https://content.historicengland.org.uk/content/heritage-counts/pub/2017/heritage-and-society-2017.pdf

- This annual report produced by Historic England is an audit of the UK's heritage assets in relation to changing aspects of society.

*Inquiry into the Civic Role of Arts Organisations. (2017). Retrieved from http://civicroleartsinquiry.gulbenkian.org.uk/

- An inquiry that raises awareness of the potential civic roles of arts organisations.

*Neelands, J. (2015). *The Warwick Commission on the Future of Cultural Value. Enriching Britain: Culture, Creativity and Growth*. Retrieved from http://www2.warwick.ac.uk/research/warwickcommission/futureculture/finalreport/

- A report from the Warwick Commission outlining how the UK might secure greater value from its cultural and creative assets.

Scottish Government. (n.d.). *Arts, Culture and Heritage Policy*. Retrieved from https://beta.gov.scot/policies/arts-culture-heritage/

- The Scottish Government's approach to supporting culture.

Speilman, A. (2017). *Health and Safety Culture in Schools*. Retrieved from https://www.gov.uk/government/speeches/ofsteds-chief-inspector-writes-about-safety-culture-in-schools

- A speech outlining OFSTED's interest in reducing the bureaucratic processes and health and safety procedures involved in out of school education in order to encourage schools to provide more diverse learning opportunities for their students.

Welsh Government. (n.d). *Arts in Education in the Schools in Wales*. Retrieved from http://gov.wales/topics/educationandskills/publications/wagreviews/arts-in-education-review/?lang=en

- The Welsh Government's approach to supporting culture.

2.3 Cultural capital and socio-economic development

2.3.1 Summary

As has been consistently shown, museum visitation has the potential to produce and reproduce what has come to be known as cultural capital, or, the social assets possessed by a person, such as language and cultural tastes, that allow them to successfully navigate a socially stratified society (Archer, Dawson, Seakins, & Wong, 2016; Bourdieu, Darbel, & Schnapper, 1991). The acquisition of cultural capital can lead to the improvement of a child's educational attainment and, eventually, to an economically and socially stable life.

Free entry to UK national museums has been a flagship cultural policy since 2001, and is, broadly speaking, a policy intended to ensure equitable access to art and culture. In addition to this, schools are able to take advantage of many other free resources, workshops and visits often offered by local and national museums and cultural institutions. However, free entry to museums is often labelled as 'false generosity' (Bourdieu et al., 1991, p. 13), since it consistently fails to improve the diversity of museum audiences.

What we can infer from this, then, is that museum access and by extension opportunities to improve academic, social and ultimately economic attainment, are not enjoyed by all children. Museum school partnerships that provide children with deep familiarity of particular museums through extended exposure could be a strategic way of addressing this problem by allowing the children least likely to visit museums with opportunities to feel at home in and take advantage of cultural spaces whilst also mitigating the risk of children from the wealthiest backgrounds disproportionately benefitting from cultural education strategies (Leroux, 2013).

2.3.2 Literatures

*Archer, L., Dawson, E., Seakins, A., & Wong, B. (2016). Disorientating, fun or meaningful? Disadvantaged families' experiences of a science museum visit. Cultural Studies of Science Education, 11(4), 917-939. doi:10.1007/s11422-015-9667-7

- A paper seeking to understand the experiences of socially disadvantaged families in museums.

*Adams, G K. (2017). Cultural learning gives children better life chances, report finds. Retrieved from http://www.museumsassociation.org/museums-journal/news/26012017-cultural-learning-better-life-chances

- This article discusses the findings of a report suggesting that cultural education can improve school attainment, particularly amongst children from low-income families.

*Bourdieu, P., Darbel, A., & Schnapper, D. (1991). The love of art: European art museums and their public. Cambridge, UK: Polity Press.

- Based on empirical data, this book is an analysis of the demographics of European art museum audiences, concluding that art museums can be sites of social exclusion and bourgeois social reproduction.

*Leroux, J.-Y., Moureau, Nathalie. (2013). Introduction to theatre at primary school: A short-term evaluation of a French experience. International Journal of Education Through Art, 9(2), 219-233. doi:https://doi.org/10.1386/eta.9.2.219_1

- This paper discusses the outcomes of a programme in a French primary school to increase students' familiarity with theatre. The programme provided extended theatre engagement opportunities for students over a one-year period. Results of a follow up survey suggested that the programme had increased the likelihood of participating students engaging with theatre independently, but that children from wealthier backgrounds were disproportionately represented since they benefited more from the programme than their less wealthy peers.

2.4 Sector literatures: museum studies and education studies

2.4.1 Summary

Museum and education literatures dealing with extended museum school partnerships can be found in many formats, reflecting the interdisciplinary nature of the topic and highlighting the close relationship between research and practice within the field. Within this body of literatures, there are two predominate areas of work relating to extended exposure to cultural experiences. The first area is centred on pedagogy, and the second around efficiency but in many cases, these issues overlap.

The successful integration of formal and informal learning practices is generally perceived to be an objective of extended museum school partnerships since it can help students to more easily relate to school disciplinary-based subjects and to understand the real-world application of the disciplines. However, there are difficulties in achieving successful integration of formal and informal learning practices based on the separation of school and museum educator communities and their differing disciplinary practices (Correia & Dillon, 2015; Pringle, 2009). To address this problem, museum school partnerships that are sustained over time, whilst initially exposing learners to both formal and informal ways of learning, could also help encourage knowledge transfer between education communities and promote the development of expertise and innovation.

It is assumed that museum visitation, as well as being able to deliver direct cognitive educational benefits, is able to deliver less tangible versions of learning such as emotional learning and the development of social skills. For example, the exposure to critical issues that some museums aim to provide might help to students to develop empathy or their own sense of identity (Hackett, 2016). Debates surrounding the evaluation of this type of museum learning persist, however, as has been shown by it seems these learning impacts are often incremental and happen over time suggesting the necessity of extended learning experiences in museums, if the full potential of museums as learning environments is to be realised.

In addition to pedagogical agendas, there is some discussion of extended museum experiences and efficiency, though this is perhaps under represented in the literatures. The pressure to be financially sustainable is acute for museums and increasingly so for schools, leading in some cases to the prioritisation of areas of work that can make direct contributions to accountability frameworks. For example, the prioritisation of literacy and numeracy in schools has coincided with the large-scale decrease in art teachers in UK schools (Neelands, 2015).

Co-location and co-production models, then, could help museums and schools tackle this problem by allowing the cost of facilities and expertise to be shared. There are several examples of co-location and co-production models in cultural organisations (some of the more relevant are included in the following case study long-list) yet this is relatively underexplored and under theorised within the museum or education literatures. However, existing research within the management sector, particularly in the development of public services (e.g. Dunleavy, 2010) and sometimes in the provision of libraries (e.g. The Labour Party, n.d.) suggest that co-location and co-production could help museums and schools to meet their cost reduction agendas.

2.4.2 Sector publications

Bellamy, K., & Oppenheim, C. (2009). *Learning to live: museums, young people and education*. London: Institute for Public Policy Research: National Museum Directors' Conference.

- This collection of essays addresses the role of cultural institutions in young people's well being and education. Contributors to the collection range from curators to academics to educators and contributions debate the need to re-evaluate the potential role of museums and other cultural institutions in holistic versions of learning.

*Bond, K. (2017). Class act: why we need schools in museums. Retrieved from http://www.museumsassociation.org/museums-journal/comment/01022017-class-act-why-we-need-schools-in-museums

- This report discusses the potential benefits of the My Primary School is at the Museum to school children, their families, museums as well as the wider museum community.

*Brown, A. (2016). Full-time learning in a museum environment. Retrieved from http://www.museumsassociation.org/comment/policy-blog/13072016-swansea-museum-my-primary-school-is-a-museum

- *This report provides first-hand support for the My Primary School is at the Museum project, particularly noting the simplicity yet potential scope of the initiative.*

Centre for the Future of Museums (2016). *Building the Future of Education Museums and the Learning Ecosystem.*

- This report discusses the future role of museums in delivering formal education, and includes a useful section addressing the problem of 'scaling up.'

*Correia, C. & Dillon, J. (2015). An evaluation of "Stronger Together": A museum learning partnership development project. Retrieved from http://www.langleyacademy.org/strongertogether/wp-content/uploads/2015/06/Stronger-Together-Evaluation-Final-Report-KCL.pdf

- This report is the final evaluation of the Stronger Together project, a project facilitating increased collaboration between teachers and museum professionals between April 2014 and April 2015. The report documents the increased opportunities for cognitive and non-cognitive learning and cross-curriculum work.

*Council for Learning Outside the Classroom (2017). Case Study: The Impact of Learning Outside the Classroom. Retrieved from http://www.lotc.org.uk/percy-main-primary-school-the-impact-of-learning-outside-the-classroom/

- This report documents the positive educational and social outcomes of a primary school student requiring additional care and educational support following an increase in the amount of opportunities to participate in learning experiences outside the classroom.

Ganley, C. (2011). Artist Rooms: Young People and Learning 2009-10. *Tate Papers, 16*.

- This paper documents Tate's Artist Rooms programme, which aims to provide authentic opportunities for young people to confidently and repeatedly engage with art.

*Martinez, F. (2017). Museums can help address empathy deficit. Retrieved from http://www.museumsassociation.org/museums-journal/comment/01062017-museums-can-help-address-empathy-deficit

- This report discusses how museums might be used as a resource to develop social skills such as empathy and to address issues of identity and difference. The report is an outcome of "Exceptional & Extraordinary", a project led by the Research Centre for Museums and Galleries at the University of Leicester

Pringle, E. (2009). The Artist as Educator: Examining Relationships between Art Practice and Pedagogy in the Gallery Context. *Tate Papers, 11*.

- This article examines the relationship between art practice and pedagogy in the context of the art gallery and suggests that the artistic processes of conceptual enquiry and meaning making can be aligned with pedagogic practice.

Steel, P. (2016). Hancock: museums must work with schools on art history A-level. Retrieved from http://www.museumsassociation.org/museums-journal/news/14122016-hancock-crucial-that-museums-work-with-schools-art-history-a-level

- This report underlines the mutually beneficial relationships schools and museums must develop and nurture in order to support the provision and realise the potential of art history education in schools.

Stephens, S. (2017). V & A unveils education programme for schools. Retrieved from http://www.museumsassociation.org/museums-journal/news/05052017-v-and-a-unveils-education-programme-for-schools

- This report documents the V&A's DesignLab programme, which is designed to support the Design Technology GCSE qualification through the provision of workshops and resources as well as partnerships.

*Sullivan, N. (2017). Educational visits to DCMS museums are decreasing. Retrieved from http://www.museumsassociation.org/museums-journal/archive-search/31012017-numbers-of-educational-visits-to-museums-on-the-decrease

- This report documents the implications of the decline in educational visits to DCMS funded museums identified through analysis of performance indicators.

2.4.4 Academic literatures

*Aerila, J.-A., Rönkkö, M.-L., & Grönman, S. (2016). Field Trip to a Historic House Museum with Preschoolers: Stories and Crafts as Tools for Cultural Heritage Education. *Visitor Studies, 19*(2), 144-155. doi:10.1080/10645578.2016.1220187

- A paper presenting the impacts of extending a field trip to a historic house museum through post-visit, creative classroom-based activities. The paper argues that writing stories in response to field trips can help children to reflect on and analyse their cultural experiences, rather than simply retain facts.

Amos, R.,& Reiss, M. (2012). The benefits of residential fieldwork for school science: Insights from a five-year initiative for inner-city students in the UK. International Journal of Science Education, 34(4) 485–511. doi:10.1080/09500693.2011.585476

- This paper evaluates data collected from 2,706 11 – 14 year old students, 70 teachers and 869 parents/carers from 46 schools in London participating in residential field trips. The paper examines the impacts of residential fieldwork and suggests that collaborative and interpersonal skills were improved as a result of residential fieldwork participation.

Dilli, R. (2016). Conducting museum education activities within the context of developing a nature culture in primary school students: MTA natural history museum example. *International Journal of Environmental & Science Education, 11*(2), 75-84.

- This paper discusses the way in which museum education strategies might be used to develop school students' knowledge and understanding of nature.

Elliott, J. (2012) To Infinity and Beyond: Museum school partnerships beyond the field trip. Unpublished Masters dissertation: Seton Hall University.

- This Masters degree dissertation examines four American museum school partnerships that offer students extended museum or cultural experiences. Using the evaluation reports of each partnership, Elliott argues that such programmes have the potential to benefit teachers, communities and individuals.

Falomo Bernarduzzi, L., Albanesi, G., & Bevilacqua, F. (2014). Museum Heroes All: The Pavia Approach to School-Science Museum Interactions. *Science & Education, 23*(4), 761-780. doi:10.1007/s11191-012-9541-x

- This paper outlines the methodology informing the Pavia approach to museum school partnerships in the context of learning science. The approach aims to alter the typical format of school trips to museums by allowing students to plan their own visits and by exploring the differences between formal and informal learning.

*Foreman-Peck, L., & Travers, K. (2013). What is distinctive about museum pedagogy and how can museums best support learning in schools? An action research

inquiry into the practice of three regional museums. *Educational Action Research, 21*(1), 28-41. doi:10.1080/09650792.2013.761924

- This paper documents the exploratory findings of an action research project asking how museum and school learning pedagogies and practices might best be integrated. The paper raises questions such as, what is the value of the museum to school education and how might museum pedagogies be used in schools?

*Hackett, A. (2016). Young Children as Wayfarers: Learning about Place by Moving Through It. *Children & Society, 30*(3), 169-179. doi:10.1111/chso.12130

- This paper documents and analyses the way in which young children develop ways of knowing museum spaces through their embodied experiences of museums and uses Ingold's theory of wayfaring as a framework. It aims to pioneer non-cognitive learning models within the museum context and is based on data collected from participants spending extended amounts of time over one year within a museum.

Hooper-Greenhill, E. (2007). *Museums and education: purpose, pedagogy, performance*. London: Routledge.

- This book argues that there is a need to theorise museum learning, particularly in a climate of accountability. The book uses empirical case studies of UK museum learning to suggest a conceptual framework based on generic learning outcomes able to effectively evaluate the impacts of museum learning.

Kelly, L. & Groundwater-Smith, S. (2009). Revisioning the physical and online museum. A partnership with the coalition of knowledge building schools. *Journal of Museum Education, 34*(4), 55–68.

- This paper outlines how the Australian Museum has worked with students over a five-year period to develop its museums, programmes and digital environment. The paper then suggests ways that museums could construct transformative learning experiences for young people.

*Kim, M., & Dopico, E. (2016). Science education through informal education. *Cultural Studies of Science Education, 11*(2), 439-445. doi:10.1007/s11422-014-9639-3

- This paper examines the ways in which the boundaries between formal and informal education can be transgressed through close collaboration and integration between school educators and museums. The authors argue that such transgressions can help science students develop as members of a problem solving community.

*Kisiel, J. F. (2014). Clarifying the complexities of school–museum interactions: Perspectives from two communities. *Journal of Research in Science Teaching, 51*(3), 342-367. doi:10.1002/tea.21129

- The authors of this paper believe that better integration of formal and informal education practices can contribute to greater STEM learning. Using a communities

of practice lens (Wenger, 1998), the authors explore the ways in which museum and school educators work together and suggest the challenges and opportunities inherent in such collaborations.

Luehmann, A.L., (2009). Students' perspectives of a science enrichment programme: Out-of-school inquiry as access. *International Journal of Science Education*, *31*(13) 1831-1855.

- This paper reports students' perspectives on an out-of-school science experience, the culmination of a one-year partnership between their school and a university science outreach programme. The paper notes how the programme positively impacted students experience and understanding of real world science as well as their understanding of subject-based science. The paper also notes that potential of these impacts for students attending under-resourced schools.

Rose, S. W. (2016). Museum–University Partnerships Transform Teenagers' Futures. *Journal of Museum Education, 41*(4), 286-292. doi:10.1080/10598650.2016.1232510

- This paper is a report of the New Bedford Whaling Museum's High School Apprenticeship programme. The paper outlines the motivations and objectives of the programme: to address the low local high school graduation rates through the affordance of structured, extended museum participation and the provision of academic administrative assistance.

Spektor-Levy, O., Aloni, Oshra, Zion, Michal (2016). "A place where nobody makes fun of me because I love science" - an in-school Mini Science Museum as a meaningful learning environment to its student trustees. *International Journal of Environmental & Science Education, 11*(18).

- This paper is a study of a mini-museum within a school environment where students act as "trustees". Through phenomenological analysis of trustee meetings, the paper suggests that the mini museum allowed students to develop cognitive, emotive and practical skills and help them feel part of a community.

*Stocklmayer, S. M., Rennie, L. J., & Gilbert, J. K. (2010). The roles of the formal and informal sectors in the provision of effective science education. *Studies in Science Education*, 46(1): 1–44. doi:10.1080/03057260903562284

- This paper summarises the weaknesses of formal science education and discusses how informal science education could tackle these weaknesses and improve science-learning outcomes.

Suter, L. E. (2014). Visiting Science Museums During Middle and High School: A Longitudinal Analysis of Student Performance in Science. *Science Education, 98*(5), 815-839. doi:10.1002/sce.2111

- This exploratory study based on longitudinal data suggests that frequent science museum attendance across the school year can contribute to increased levels of cognitive educational achievement.

Thomson, P., & Clifton, J. (2013). Connecting with Parents and the Community in an Urban Primary School: Creative Partnerships to Build Literacy/ies. In *International Handbook of Research on Children's Literacy, Learning, and Culture* (pp. 54-66): London. Wiley.

- This chapter documents and analyses the use of one primary school's extended use of creative practitioners within classrooms to deepen connections with students' parents.

Tran, N. A. (2011). The relationship between students' connections to out-of-school experiences and factors associated with science learning. *International Journal of Science Education, 33*(12), 1625-1651.

- This paper examines and compares how students and teachers associated out of school experiences with learning.

*Watermeyer, R. (2015). Science engagement at the museum school: teacher perspectives on the contribution of museum pedagogy to science teaching. *British Educational Research Journal, 41*(5), 886-905. doi:10.1002/berj.3173

- This paper explores museum-school teachers' accounts of object-based learning. In doing so, the author considers the potential of object-based learning in harmonising informal and formal types of learning and allowing students the ability to connect school learning to their everyday lives.

3. Case studies

3.1 Summary

This section of the report lists and describes programmes that relate to the My Primary School is at the Museum project. Whilst the format of some of the programmes included in the list resemble the residency model of My Primary School is at the Museum, other formats differ. Despite this, all programmes listed are focused on the delivery of formal learning within a cultural setting over an extended period of time. The final section of the list includes programmes developed and delivered in the US.

3.2 Case studies

Activity	Description
My Primary School is at the Museum Swansea, Tyne and Wear, Liverpool 2016	*My Primary School is at the Museum* tested the hypothesis that there may be beneficial learning, social and cultural outcomes for primary school children and their families when a significant portion of their learning takes place in a museum setting, as well as demonstrating the benefits for museums. Children from two primary schools and a nursery, from Tyne & Wear, Swansea and Liverpool, were based at their local museum for up to a term between January and June 2016, receiving their normal school education whilst in the museums. It aimed to create and evaluate a potentially symbiotic relationship between primary schools and museums that could develop into a new model of educational delivery. Such a hybrid model could potentially help to resolve the nation's shortage of primary school places, while simultaneously supporting resilient and sustainable operating principles for our museums. The museums' collections were used to provide context for a range of school subjects. Facilities at the museums were arranged to enable children to absorb these local collections directly and indirectly with constant connections being made between objects and the curriculum. Throughout the project, teaching followed primary education best practice and continued to deliver the requirements of the National Curriculum and the Early Years Foundation Stage statutory framework. The museum hosted the school's day-to-day programme of lessons, including lunches and breaks, and offered a fundamentally different experience from the more usual occasional one-day visits. The pilot project suggests that it is important to develop an evidence base for the development of extended museum school partnerships, particularly in respect of the evaluation of educational, social and community impacts of partnerships. However, evaluation of the pilot project highlights some areas for consideration in future work and development, including, the flexibility needed for partnerships to be successful, the emphasis on child-centred learning and community involvement and children's improved communication skills.

Class in Residence Captain Cook Birthplace Museum and Stewart Park October 2017	This is a week long “residency” by a Year 1 class at the Captain Cook Birthplace Museum and Stewart Park planned for October 2017. The project was inspired by the My Primary School is at the Museum project and is a development of the Tees Valley Making a Mark programme (listed below) which afforded school children the opportunity to visit the same museum at least three times over an academic year.
My Nursery School is at the Museum and Garden The Fitzwilliam Museum and Cambridge University Botanic Garden October 2017	Based on the My Primary School is at the Museum, this project will run in October 2017 and see children from a nursery school based at the Fitzwilliam Museum and the Cambridge University Botanic Garden. Children will spend two days at each site and their final day will be spent at the site of their preference.

The Langley Academy	The Langley Academy, established in 2008, is a secondary school in Slough and the only museum learning school in the UK, inspired by models found in the US. The museum learning framework used by the school is embedded in all learning practices and is based on the principles of object-led curiosity, exploration and discovery. The school has multiple partnerships with local and national museums and aims to introduce museum learning to the primary schools in its cluster. As well as integrating museum learning across all curriculum subjects, the school has its own on-site exhibition space, where children can curate temporary exhibitions, and a museum club, both of which contribute to children's understanding of museum practices. The Langley Academy is partnered with three local museums and one national museum: The Museum of English Rural Life, The Oxford University Museum of Natural History, The River and Rowing Museum and the Victoria and Albert Museum. The museums form the Museums Advisory Group to the Langley Academy, offering support and guidance for the strategic development of the school's museum learning programme. In addition, the partner museums often operate as the museum contexts in which learning is delivered at the Langley Academy. The Langley Academy view museums as gateways to real things, real stories and real people and utilise museum collections to make learning meaningful and memorable. They believe that museums and their collections develop students': • Love for learning • Curiosity and creativity • Ideas and critical thinking • Self-esteem and identity • Communication skills • Understanding of context and links • Knowledge of the past • Hopes for their future Jenny Blay, Head of Museum Learning, advises – "Our aim at Langley is to achieve better outcomes for students using museums. We have around 1200 students and 120 teachers with a take up level of over 10,000 student places on Museum Learning activities each year. We don't currently have plans to work towards accreditation and don't really have a collection of our own other than an assortment of handling objects. We do have some stunning museum objects on long-term loan and then an exhibition programme that changes roughly six

Cheney School	Cheney School is partnered with and hosts the East Oxford Community Classics Centre, which is working towards the museum accreditation standard. Students at the school engage with the centre in a number of ways, including projects on Roman mosaics and ancient cookery. Additionally, children are working with museum staff to train as curators as part of the museum accreditation standards process. The East Oxford Community Classics Centre has a museum collection, including Roman pottery, weaponry and glass, Greek and Roman coins on loan and some lithic and medieval items, many of which are objects that are integrated into year seven history lessons at the school. The centre also welcomes visitors and possesses a large lending library and a variety of replica artefacts, as well as running a number of community festivals and themed days. Like the Langley Academy, Cheney School is focused on academic excellence, with the museum school aspect of the institution playing a supporting role.

Stronger Together 2014-2015	Stronger Together facilitated 11 extended partnerships between schools and museums during 2014 – 15 and was led by the Langley Academy and The River and Rowing Museum. The partnerships allowed teachers and museum practitioners to develop collaborative learning projects with curriculum, and sometimes cross curriculum, links. Children participating in the programmes were expected to gain increased curriculum knowledge and understanding, as well as other skills such as communication or reasoning. One example of a partnership consisted of a group of 14 year seven students from an Oxfordshire school attending a launch event, 9 after-school sessions and two sessions at the partner museum. The mixed-ability group focused on developing their describe, analyse, interpret, value thinking skills through the lens of history and art and design. In addition, the partnership aimed to help participants increase and improve their digital skills through the use of iPads and different software packages. The project evaluation suggests that as well as increasing their subject knowledge, children participating in programmes began to develop an increased sense of pride in their school work. For teachers and museum professionals, the project afforded greater opportunities for collaboration, and educators from both communities reported gaining deeper understanding of other and new ways of working, which they could respond to in their own work during and beyond the project. Despite the administrative burden and reliance on staff enthusiasm that is highlighted in the project evaluation as being necessary for the success of extended school museum partnerships, the programme might also be seen as scalable in the sense that it effectively facilitated the sharing of expertise. Not only did the Stronger Together programme afford Langley Academy (a museum learning school) the opportunity to share their way of working with other schools and museums, but teachers and museum educators were given the tools and time to reflect on their own and each others practices, resulting in collaborative professional development. In essence, then, though relatively small numbers of children participated in programmes, the impacts have the potential to extend beyond the initial partnerships and their direct participants.

Associate Schools Programme The Southbank Centre On going	The Southbank Centre works with 10 primary schools from their local borough. The primary schools are known as Associate Schools and are invited to produce and attend special events at the Southbank centre related to the institution's current exhibitions and programmes. Events happen during term time and out of term time meaning that the families of students can be involved in the programme too, helping to ensure the Southbank Centre is a locally engaged institution.
Art, opera and poetry Quest Ongoing Children and Arts	Quests take children who are least likely to participate in cultural activities on year-long cultural journeys; 25,000 children have participated in quests since they were established in 2008. All quests commence with participating children visiting their local cultural venue, which might be a theatre, opera house, museum or gallery. After their visit, children participate in workshops with professional practitioners (artists, storytellers, singers, musicians), giving them an opportunity to explore their own creativity. After more workshops at school or in the partner institution, each quest culminates in an opportunity for children to exhibit their work in their partner gallery or perform on stage at their partner venue. During the quests, there is no focus on the delivery of the formal curriculum, instead the programme is focused on developing creativity, providing equitable access to the arts and increasing children's knowledge and understanding of professionalised visual and live arts. Children and Arts is a charitable organisation and notes that for many of the children taking part in their programmes, it is their first experience of visual or live arts in a professional setting.

Schools and Writers First Story 2016-2017	During the academic year 2016-17, the Schools and Writers programme established over 70 writing residencies in UK schools. Published writers are matched with a school or class and visit once a week, usually for six weeks, each time delivering a writing workshop. The Schools and Writers programme has two aims; first, to encourage creative writing amongst children, and second, to provide stable incomes for published writers. All teachers report that children's writing improves following residencies, and three quarters of children see this improvement in themselves. Other measures of impact include confidence and creativity and raised aspirations. Children participating in the programme take part in weekly workshops, developed and delivered by their writer in residence, and produce different types of creative writing over the lifespan of the residency, including poetry, fiction and scripts. As well as improving writing, the workshops can focus on improving children's speaking and listening and thus help children to develop a broad range of skills that are useful in school contexts as well as non-school contexts. Evaluation of the impacts of the Schools and Writers programme has shown benefits to students' creativity, communication, confidence and resilience. Schools and Writers is run by the charitable organisation First Story. First Story work with schools where 50% of students or more are deprived according to the Income Deprivation Affecting Children Index and/or GCSE results fall in the lowest third of the national distribution. This strategy aims to address the inequalities between privately educated children and children educated in non-fee paying schools through the provision of robust and enduring cultural education. As a particularly scalable extended cultural education programme, Schools and Writers underlines some of the key issues facing the development of such partnerships. As we have seen, the Schools and Writers programme is national but also local, since each writer tailors each residency to the needs of their school. In addition, since the programme takes place within schools during the existing school day, there is no additional travel requirement for children, and limited additional administration, especially when compared to the administration associated with school trips. Children and their schools benefit since they are afforded a unique extended culture experience with reduced logistical and administrative burdens.

Young Curators On going The Wallace Collection	The Young Curators programme evolved out of a project in 2009-10, which developed an exhibition curated by children. This was a partnership project with St. Vincent's RC Primary School. For the exhibition, 12 children in years five and six were responsible for choosing the theme, content and design and writing the panel text. After the exhibition project, the partnership built on the child-led tours to the public that had evolved spontaneously during the exhibition. There are now 10-12 children in the team from years four, five and six. They are trained in presentation and public speaking (for tours, which are delivered three times a year). Each tour follows the process used for the exhibition: the children choose an object, research it and write a script. Children also spend time talking to curators, conservation staff and the press team to gain an insight into the work of the Wallace Collection. Young Curators at the Wallace Collection do not receive curriculum education within the museum but report learning about practical museum work, how to research and think about objects and consistently report increases in confidence when communicating with audiences and individuals. The programme is also perceived by children as a privilege, or as a reward for good behaviour and work in school. Children usually remain part of the programme for three years. In response to the popularity of the programme, the Wallace Collection organically established Wallace Youth, a programme for secondary school aged children who had completed the Young Curators programme but wished to remain connected to the institution.

Making a Mark Tees Valley Museums, schools and National Portrait Gallery Ongoing	Making a Mark assists schools in curriculum delivery by developing museum education programmes that link to one or more parts of the school curriculum. Children attend at least three visits to museums within the partnership during an academic year. In addition to curriculum delivery, the programme encourages children to understand and be proud of their local heritage, identity and mark on society, and in doing so, aims to raise aspirations. Making a Mark is a partnership project between eight Tees Valley museums and the National Portrait Gallery. It is part of the Museums and Schools Programme 2012–16 funded by Department for Education. The programme enables regional museums to develop educational opportunities for local schools by working closely with national museums. It aims to increase the number of high-quality educational visits by schools in areas that currently have lower than average cultural engagement. This work is closely supported by the Arts Council and regional bridge organisations. Making a Mark has worked with over 280 schools and delivered over 40,000 visits across the partner museums. In the first three years, each class that took part visited for three sessions, enabling a deeper relationship to develop and for a greater impact on students' work. Since April 2015, schools have been free to choose how many sessions students participate in, but Making a Mark encourages the idea of at least two led sessions for schools using the programme to offer high quality, active and creative learning experiences. Evaluation has shown that almost all teachers taking part feel that the programme has made a real difference to students' interest in their school subjects and raised their awareness of their local heritage. Encouragingly, strong feedback from teachers also indicated that Making a Mark had directly impacted on the quality of students' work.

Family Collective Tate Liverpool	Family Collective is an artist-led programme run by Tate Liverpool. The programme establishes relationships with mothers of school age children facing disadvantage through a local children's centre. Family Collective meet at Tate Liverpool once a week during school hours and co-produce Tate's successful half term family workshops. Once a programme has been designed, Family Collective meets at a weekend so that the members can bring their children to test the workshop. Family Collective aims to engage with mothers facing disadvantage, explicitly addressing socio-economic disadvantage, empowering them to access free services offered across Liverpool particularly education and healthcare. By helping mothers facing disadvantage to access education and healthcare, the programme hopes that the children of Family Collective members will also benefit. Two members of Family Collective now study art at college and several teen-aged children of Family Collective members have completed work experience placements at Tate Liverpool.
DesignLab Victoria and Albert Museum On going	DesignLab is a programme of special events available for state-funded schools. Students and teachers participating in DesignLab are exposed to contemporary and historical design, design practitioners, and museum collections in an in depth way to promote understanding of creative careers and connect with learning across the school curriculum, particularly with STEAM subjects.

School Without Walls The Egg, Theatre Royal Bath Ongoing	The School Without Walls programme began as a small partnership between Egg, the children's theatre of the Theatre Royal Bath, 5x5x5=creativity an independent arts-based action research organisation which supports children in their exploration and expression of ideas, helping them develop creative skills for life, and St Andrew's C of E Primary School. In the summer term of 2011, a class of 25 students spent seven weeks during the summer term at the Egg, this was repeated by a second class of 25 students the following summer. Through each residency, the theatre became the students' classroom. Students were given open access to 5x5x5=creativity artists, egg performances, theatre staff and the city at large and as well as having a teacher, has a lead artist too. The project transformed the egg's relationship with the participating school and also transformed the school's attitude towards external venues as learning environments. Following this pilot programme, School Without Walls entered a second phase, expanding its provision across other schools, including a secondary school. This phase of the School Without Walls project aimed to demonstrate how arts organisations can drive measurable change in schools' approaches to creativity and culture in the curriculum, engagement with the arts, learning outside the classroom and pupil attainment. The overall intention was to develop a new model of learning that enables cultural organisations to link to the education system in a different way and achieve better learning outcomes for all.

Heritage Schools 2012 - 2018	The Heritage Schools programme plays a key role in delivering the UK Government's cultural education policy. The large-scale and well-established programme operates through a network of 8 regional centres, managed by Historic England, the public body with statutory responsibility for the protection of the historic environment. Each centre is staffed by a Local Heritage Education Manager (LHEMs) who works with up to 24 schools. The aim of Heritage Schools is to help school children develop an understanding of their local heritage and its significance. It achieves this through the provision of CPD training opportunities for qualified teachers and Newly Qualified Teachers (NQTs) as well as training for those enrolled on Initial Teacher Training Courses. In addition to CPD training, Heritage Schools provides teaching resources, and LHEMs broker and facilitate relationships between schools and local heritage providers, signposting and unlocking access to existing heritage resources such as religious sites, historic buildings and libraries and archives.
Eureka Nursery School Eureka: The National Children's Museum On going	The Nursery School at Eureka is a private nursery for children aged 0 - 5 situated within the museum and run by museum staff. During school holidays, childcare is expanded to provide care for older children too. Children at Eureka nursery spend time in the museum, using resources to learn and socialise.
Baltic Street Adventure Playground Established 2013	A child-led adventure playground based in Glasgow. The initial concept was designed and delivered by Turner Prize-winning architecture collective, Assemble. The playground provides supervised and free out-of-school care for 6 – 12 year old children. The child-led approach to play encourages children to make independent decisions, manage risk sensibly, become leaders, and work effectively within a team. Baltic Street Adventure Playground aims to; afford children their right to play free from undue adult pressures, be physically, emotionally and socially nurturing as well as to promote inclusivity. The organisation aims to increase the age range it serves within the next ten years.

Summer School Tate Annually	Summer School is an extended continuing professional development (CDP) course for teachers, which runs annually during the school summer holidays. As well as providing development opportunities framed by artistic practices for participants, Summer School convenors use the process and outcomes of the programme to inform the production of Tate resources for schools and teachers. "Summer School is framed by artistic practice, where it intersects with teaching and connects to Tate's collection and exhibitions.
Archaeological Science at Flag Fen The Natural History Museum and Thomas Deacon Academy 2016 – onwards	This practitioner-led pilot programme offered KS3 students from one school an opportunity to experience real-world science investigations and to meet science professionals, whilst meeting the requirements of their formal science learning. The project continues and aims to deepen its partnership with the school by delivering more science education in heritage settings.
US and international Museum Schools	There are several museum schools in the US, including: http://www.nycmuseumschool.org https://www.bankstreet.edu http://www.brentelementary.org/ In addition, there is one museum school in India: http://parvarish.weebly.com/ The New York City Museum School inspired the founding of the UK's only museum school, the Langley Academy. Notably, the Museum School, Bhopal, India was founded on principles of educational and social equity.
Studio Classroom Programme Ogden Museum of Southern Art, New Orleans On going	In the Studio Classroom Program, school children move to the museum for a week of intensive arts experiences and immersion in the museum environment, mixed with traditional subject lessons in the afternoon.
School Partnerships The Whitney Museum of American Art, New York On going	The programme creates long-term, multi-year partnerships with a small group of New York City schools that focus on the whole school community: students, their families, teachers, and administrators.

High School Apprenticeship The New Bedford Whaling Museum	The High School Apprenticeship scheme run annually by the New Bedford Whaling Museum, US was established to help address the low rate of local high school students making college applications. The programme provides academically motivated high school students from low-income backgrounds with resources and experiences that promote community engagement and personal and academic development. Additionally, the programme provides practical assistance with college and financial assistance applications. The programme requires sustained commitment from apprentices since it is three years long and apprentices are expected to attend Monday-Thursday throughout July and August and Monday-Thursday after school during term time.

4. Insights and recommendations

This section of the report compares and contrasts the outcomes of the different programmes included in the case study list, drawing together and analysing the impacts of formal curriculum delivery within museum settings over extended periods of time to present a series of insights and recommendations for the development of My Primary School is at the Museum.

4.1 Curriculum optimisation

Successful curriculum delivery is a key concern of extended museum school partnerships and one which could be easily evaluated through the use of existing assessment frameworks. The national examination results of children at the Langely Academy, who are exposed to object-based learning in their everyday school lives, for example, demonstrate that extended museum school partnerships are capable of successfully delivering the curriculum. Teachers assessments of children participating in the My Primary School is at the Museum project and Making a Mark also indicate successful curriculum delivery within museum contexts.

It seems, then, that there is less need to question whether museums are able to satisfy curriculum needs. More pressing is to question what additional impacts result from curriculum delivery through extended museum school partnerships, or how extended museum school partnerships are able to optimise curriculum delivery. It should also be remembered that though schools and museums, it seems, are both suitable contexts for curriculum delivery, it is not the case that all children thrive in schools. For some children, perhaps, museums may in fact be a preferable context for successful curriculum delivery.

4.2 Applied understanding

In several of the case studies, the children participating in extended museum school partnerships developed applied understandings of subjects that impacted them in different ways.

For children at the Langley Academy, object-based learning is reported to afford disciplinary subjects a real-world context that can help link subjects to home life, other school subjects and career paths, positively impacting cognitive subject-based learning and increasing career choices.

4.3 Other learning

Within most of the case studies there is a focus on the non-cognitive learning opportunities arising from extended museum school partnerships.

Improved communication is widely reported as a key outcome for children participating in extended museum school partnerships, suggesting that museums (or

other cultural settings) are more effective at supporting communication development than traditional classrooms. Some programmes, such as the Wallace Collection's Young Curators, explicitly focus on improving children's public speaking skills whilst others, such as Baltic Street Playground, are more implicit at developing communication, encouraging children to listen and respond thoughtfully to contribute effectively to the playground community.

Other programmes also report the improvement seen in children's communication skills during extended museum school partnerships, despite no more attention being paid to developing communication skills than would be paid in a normal classroom setting. Nursery children participating in the My Primary School is at the Museum Project, for example, increased their communication circles and primary school aged children participating in the same programme were able to lead a tour of their museum long after the end of the programme, demonstrating strong knowledge retention and confidence in their own ability to share their knowledge.

4.4 Participant attitudes

Perhaps one of the most interesting impacts of extended museum school partnerships is the difference they can make to participants' attitudes to learning. The Stronger Together evaluation report suggests that children felt more pride in their school learning than they had done before participating in programmes, a feeling replicated by children participating in Young Curators, who felt privileged to be involved in the programme. This impact can also be attributed to children participating in My Primary School is at the Museum, who, generally speaking, had an improved attendance rate and in one case a class recorded their best attendance for the duration of the programme. Impacting participant attitudes to learning and engagement is perhaps a feature of extended museum school partnerships that is shared with museum outreach work, something underlined by Family Collective where participants are encouraged to feel empowered.

4.5 Resource management

Though we might assume that co-locating schools and museums might reduce costs, there is little practical evidence to support this within the sector. Other sectors, however, might offer translatable findings or evaluative frameworks that could effectively assess the financial benefits of co-locating schools and museums.

In fact, extended museum school partnerships, it seems, demand additional administration resources and logistical planning which may increase costs. The Stronger Together partnership suggests that this means the success of extended museum school partnerships can sometimes rely on the enthusiasm and commitment of teaching staff and museum learning practitioners. In some case studies though, the threat of increased administration and logistics is mitigated or reduced through strategic planning.

The Tyne and Wear-based My Primary School is at the Museum programme, for example, matched a primary school with a museum located on the other side of the

road. The Cheney School, as well, houses its partner museum within its walls. The proximity of these partner schools and museums ensures minimal additional travel for participants, reducing the time and cost of travel, as well as its administrative burden. Of course, not all schools are located close to a museum, but other cases demonstrate different ways of reducing the need for additional administrative and logistical resource.

Risk assessments are perceived as a barrier to extended museum school partnerships, since children are removed from the relative safety and security provided by schools. Baltic Street Adventure Playground takes an innovative approach to the problem of risk assessment by building risk awareness into its learning framework, an approach supported by OFSTED. This type of approach acknowledges the presence of risk and helps children learn how to manage themselves and others in risky situations. Though this approach may remove some additional administrative resource, resource is still required to develop and deliver appropriate learning frameworks, though arguably the benefits to the children could make this worthwhile.

Schools and Writers, a residency model whereby writers rather than children take up a new residency, is perhaps the most effective way of reducing the additional administrative resource associated with extended museum school partnerships, since children's normal school day is not disrupted by travel.

In addition, the case studies highlight the both the commercial and fundraising viability of extended museum school partnerships. The Eureka Nursery, for example, is a fee-paying nursery, to which its museum partner lends a unique, and exciting, selling point. Several of the case studies rely on fundraised income, raised through individuals as well as grant-making bodies. Both funding models suggest the potential for developing extended museum school partnerships without relying on existing school or museum budgets.

4.6 Recommendations

The final section of this paper makes general and specific recommendations in response to the impacts of extended museum school partnerships identified in this report and that could help optimise the potential of such programmes.

1. Collect demographic data in conjunction with learning outcomes and attitudes to learning data from historic (where possible) and future programmes to analyse the socio-economic impact of extended museum school partnerships.

Many programmes featured on the case study list are orientated, implicitly or explicitly, towards children from socio-economically deprived backgrounds. The proportion of children receiving free school meals, examination results, or whose first language is not English are often used as a demographic indicators. However, this data indicates only school-wide proportions of deprivation and there could be a risk that children from wealthier backgrounds disproportionately benefit from extended museum school partnerships, since they are generally better equipped to access such programmes. Collection of more detailed and and/or longitudinal data could explore this risk and allow comparison of extended museum school partnerships with other

strategic socio-economic interventions such as community outreach programmes or free entry policies, allowing resources to be concentrated in the most beneficial areas.

2. Examine the role of extended museum school partnerships in developing children's communication skills.

Communication, whether an explicit feature of a programme's learning framework or not, has been identified as a key area of learning for children participating in extended museum school partnerships. Further research focused on communication skills development in this area could emphasise how the museum is a context that can optimise curriculum delivery.

3. Develop a strategic framework for establishing successful extended museum school partnerships.

The case studies suggest that certain factors influence the success of extended museum school partnerships, for example ease of travel between sites, administrative capacity and staff commitment and enthusiasm. Developing a framework for establishing partnerships could help interested schools, museums, teachers and museum practitioners work with the right partners in a sustainable way from the outset.

4. Identify scalable or transferable aspects of extended museum school partnerships

Many partnerships are bespoke and though this means children participate in unique and tailored programmes, it could be difficult to replicate or grow partnerships in different contexts. However, some partnerships are scalable and adaptable and could offer useful insights into developing larger partnership programmes. The professional development opportunities afforded by some of the programmes included in the list of case studies, for example, mean programmes are scalable in one sense, since children not participating in the programme might indirectly benefit through their teachers' improved or developed practice.

5. Explore the feasibility of the various funding or resourcing models of museum school partnerships, particularly focusing on administrative resource management, co-location and fundraising initiatives.

It is assumed the museum school partnerships may be financially beneficial to schools and museums, since they rely on shared resources and facilities. However, there is little evidence to support this hypothesis and, in fact, there is evidence to show that museum school partnerships require additional administrative resource, at least at the outset. However, there is some evidence to suggest that museum school partnerships have fundraising (particularly where programmes aim to provide equitable access to the arts) and in some cases commercial potential. Further research could focus on developing strategic management models aimed at reducing the cost of museum school partnerships.

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